

Carbon Pricing and Co-Abatement of Toxic Emissions

Evidence from Alberta's SGER/TIER Regulation

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01 The Question

Carbon prices are designed to cut CO₂ — but do they also **reduce toxic co-pollutants**?

Mechanism: A carbon price raises the cost of fossil-fuel combustion. Firms respond by cutting output, switching fuels, or adopting cleaner technology. Each channel also reduces *co-emitted* toxic substances — mercury, arsenic, lead, VOCs.

US evidence (cap-and-trade)

Pham & Roach (2024): RGGI cut coal-utility toxic releases by **62%**

Canadian question (intensity standard)

Does Alberta's SGER/TIER produce **comparable co-abatement**?

02 US Benchmark: Pham & Roach (2024)

Setting RGGI cap-and-trade, 11 NE states

Data US Toxics Release Inventory (TRI)

Sample Coal electric utilities, 2000–2019

N 4,831 facility-years, 243 facilities

Design: Two-period DiD

- Facility, state, and year FEs
- SE clustered by facility
- Reference: non-RGGI states

Results

Phase 1 (2009)	−45.4%
Phase 2 (2014)	−30.8%
Combined	−62.2%

Mechanism: hard cap forces output cuts → less coal ash, fewer stack emissions

03 Alberta's Carbon Policy

Period	Regulation	Rate (CAD/t)
2007–2017	SGER	\$15–30
2018–2019	CCIR	\$20–30
2020–2022	TIER	\$30–50

Coverage: facilities $\geq 100,000$ t CO₂e/yr

RGGI vs. SGER/TIER

RGGI: hard cap on total tonnes — quantity instrument

SGER/TIER: intensity standard — firms can grow output if intensity falls

Key: intensity standard weakens production-cut incentive → expect smaller co-abatement

04 Data

Source National Pollutant Release
Inventory
(NPRI), ECCC, 2000–2022

Industry filter 5 energy NAICS codes:
211110/211113/211114 — oil & gas extraction
221112 — fossil-fuel power generation
324110 — petroleum refineries

Excluded British Columbia (carbon tax
2008
contaminates control group)

Sample

Observations	64,316
Facilities	12,710
AB (treated)	9,895
Control prov	2,815
Years	2000–2022

Outcomes:

$$lsumemission = \log(1 + R_{it}^{air})$$

$$lsumreleases = \log(1 + R_{it}^{all})$$

Two-way fixed effects DiD:

$$y_{it} = \underbrace{\beta_1(Alberta_i \times SGER_t)}_{\text{Phase 1 effect}} + \underbrace{\beta_2(Alberta_i \times TIER_t)}_{\text{Phase 2 effect}} + \mathbf{X}'_{it}\boldsymbol{\gamma} + \alpha_i + \delta_t + \varepsilon_{it}$$

$Alberta_i = 1$ if facility in Alberta

$SGER_t = 1[t \geq 2007]$

$TIER_t = 1[t \geq 2020]$

$\mathbf{X}_{it}$: log GDP, fossil-fuel share, coal/gas price

α_i : facility FE δ_t : year FE

SE clustered by facility

Parallel trends test

Event study (ref = 2006)

Pre-trend joint Wald test:

$F(3) = 1.70, p = 0.166$

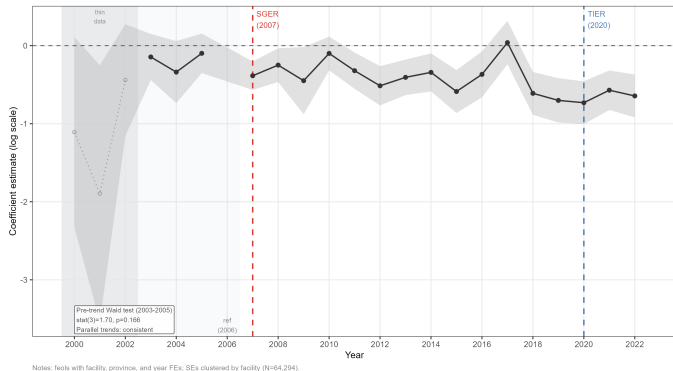
✓ **Fail to reject — consistent**

06 Results: US vs. Canada

	<i>Log Air Emissions</i>		<i>Log Total Releases</i>	
	<i>US (RGGI)</i>	<i>Canada (SGER/TIER)</i>	<i>US (RGGI)</i>	<i>Canada (SGER/TIER)</i>
$\hat{\beta}_1$ (Phase 1)	-0.605** (0.280)	-0.193*** (0.064)	-0.870*** (0.308)	-0.194*** (0.063)
$\hat{\beta}_2$ (Phase 2)	-0.369 (0.308)	-0.308*** (0.068)	-0.949*** (0.398)	-0.306*** (0.068)
<i>Begin %</i>	-45.4%	-17.5%	-56.7%	-17.6%
<i>Lower %</i>	-30.8%	-26.5%	-61.7%	-26.4%
Total %	-62.2%	-39.4%	-83.4%	-39.4%

SE clustered by facility. Bootstrap 95% CI for $\hat{\beta}_1$: $[-0.31, -0.06]$; for $\hat{\beta}_2$: $[-0.42, -0.18]$. ** $p < 0.05$,

*** $p < 0.01$.



2003–2005: flat near zero

Pre-trends hold ✓

2007: immediate drop at SGER introduction

2020: further decline at TIER introduction

2022: -0.64 log pts ($\approx -47\%$)

Open circles (2000–2002): < 120 AB obs

Canadian oil & gas sector:

99.2% of NPRI releases
are to **air**

Mean air: 384,000 kg/facility-yr

Mean non-air: 1,655 kg/facility-yr

$\text{cor}(l_{\text{summission}}, l_{\text{sumreleases}}) =$
0.999

US coal utilities:

Coal ash $\rightarrow$ land disposal

Scrubber slurry $\rightarrow$ water

So $l_{\text{sumreleases}} \gg l_{\text{summission}}$

Implication:

In Canada, the mechanism is
output curtailment, not
end-of-pipe or media substitution.

Fewer barrels/m³ produced

$\rightarrow$ less venting/flaring

$\rightarrow$ lower toxic air emissions

▶ **Co-abatement exists under intensity pricing.**

SGER still reduced toxic air emissions by **17.5%** — sufficient to shift firm technology.

▶ **Tighter prices deliver larger gains.**

TIER's higher price added a further **26.5%**. Price level matters, not just instrument type.

▶ **But cap-and-trade delivers more.**

Canada's **-39%** is half the US RGGI **-62%** benchmark. A hard cap removes the output-expansion escape valve.

As Canada's carbon price escalates toward \$170/tCO₂e by 2030, further co-abatement can be expected — these health co-benefits should enter policy benefit-cost analyses.

References

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Statistics Canada (2024). Tables 36-10-0222-01, 25-10-0029-01, 18-10-0268-01, 18-10-0004-01.

Full replication code:

`github.com/oluwafemiojo/`

`Carbon-Pricing-and-Co-Abatement-of-Toxic-Emissions-Evidence-from-Alberta`